CVPR 2026 HIGHLIGHT

# VISTA4D

## Video Reshooting with 4D-Grounded Point Clouds

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# The Precision Challenge

Why existing video reshooting methods fail to maintain 4D consistency under extreme camera shifts.

# The Reshooting Problem

## Implicit Weakness

Latent-conditioned diffusion models provide photorealism but often suffer from **Imprecise Camera Following**. Without geometric signals, identity drifts during large shifts.

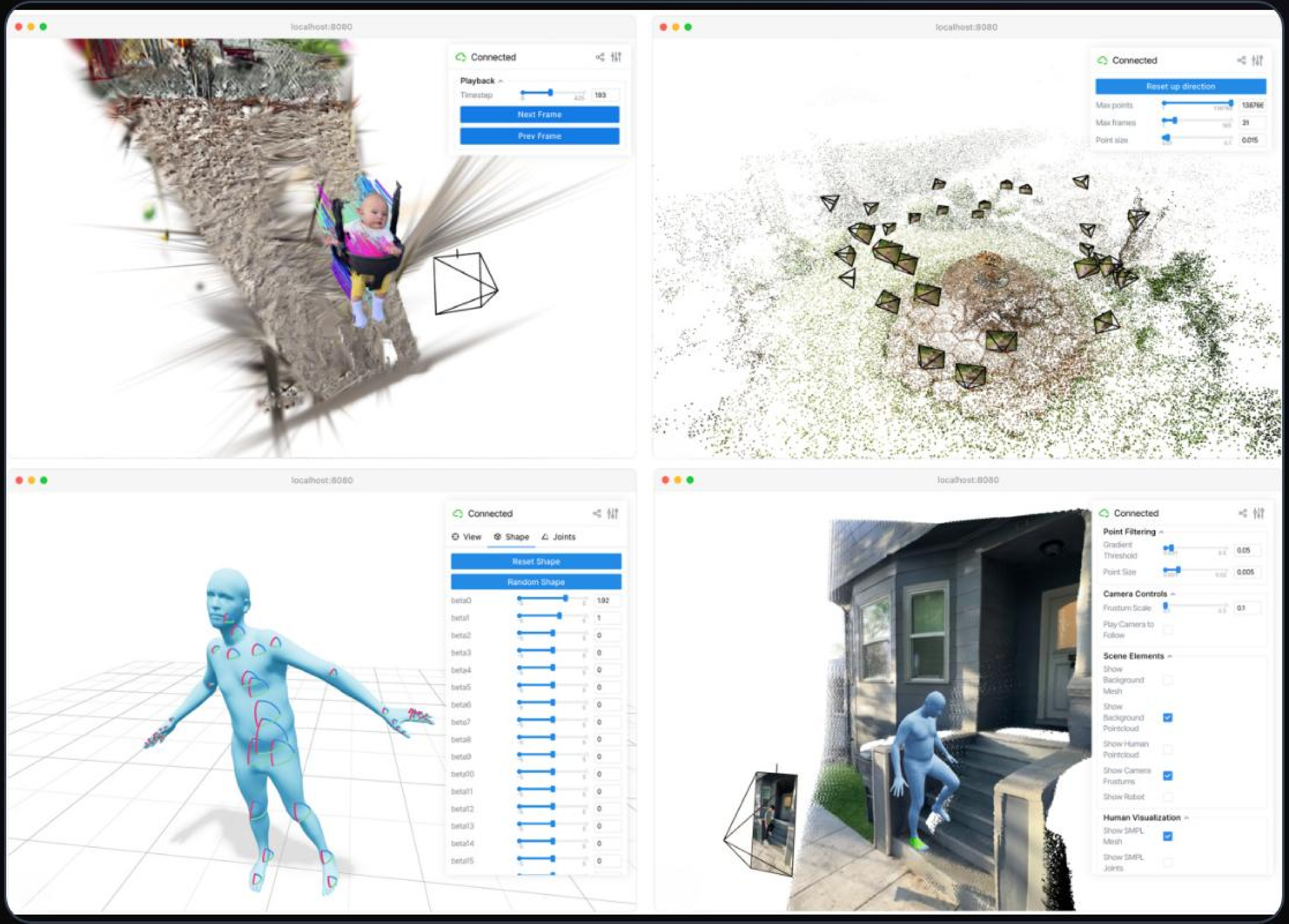
## Explicit Weakness

Prior point-cloud methods offer strong control but propagate **Depth Artifacts**. Imperfect reconstructions lead to visual glitches and "streaks" in the final output.

# The Vista4D Solution

Vista4D bridges the gap by grounding the input video and target cameras in an **editable 4D point cloud**.

- Temporally-persistent static pixels provide a stable spatial anchor.
- Explicit geometry signal matches target trajectories exactly.
- Learns to correct reconstruction noise via artifact-tolerant training.



# Core: Temporal Persistence

100%

GEOMETRIC CONTINUITY

## Static Pixel Persistence

Unlike standard 3D point clouds that refresh every frame, Vista4D lifting is **persistent**. Background pixels are accumulated across the entire video, ensuring that unseen regions carry dense geometric context as the camera moves.

# Noisy-Render Training

Real-world 4D reconstruction is inherently imperfect. Vista4D is trained on **artifact-laden renders** rather than "perfect" ground truth.

This "Noisy-Render" regime teaches the model to treat the point cloud as a **guide**, using its generative priors to hallucinate photorealistic lighting and geometry where the signal is weak.

# System Architecture

### Joint Latents

Concatenates source tokens and target point-cloud renders along the frame dimension.

### Plücker Rays

Camera parameters are injected as Plücker embeddings at each transformer block.

### Wan2.1 Base

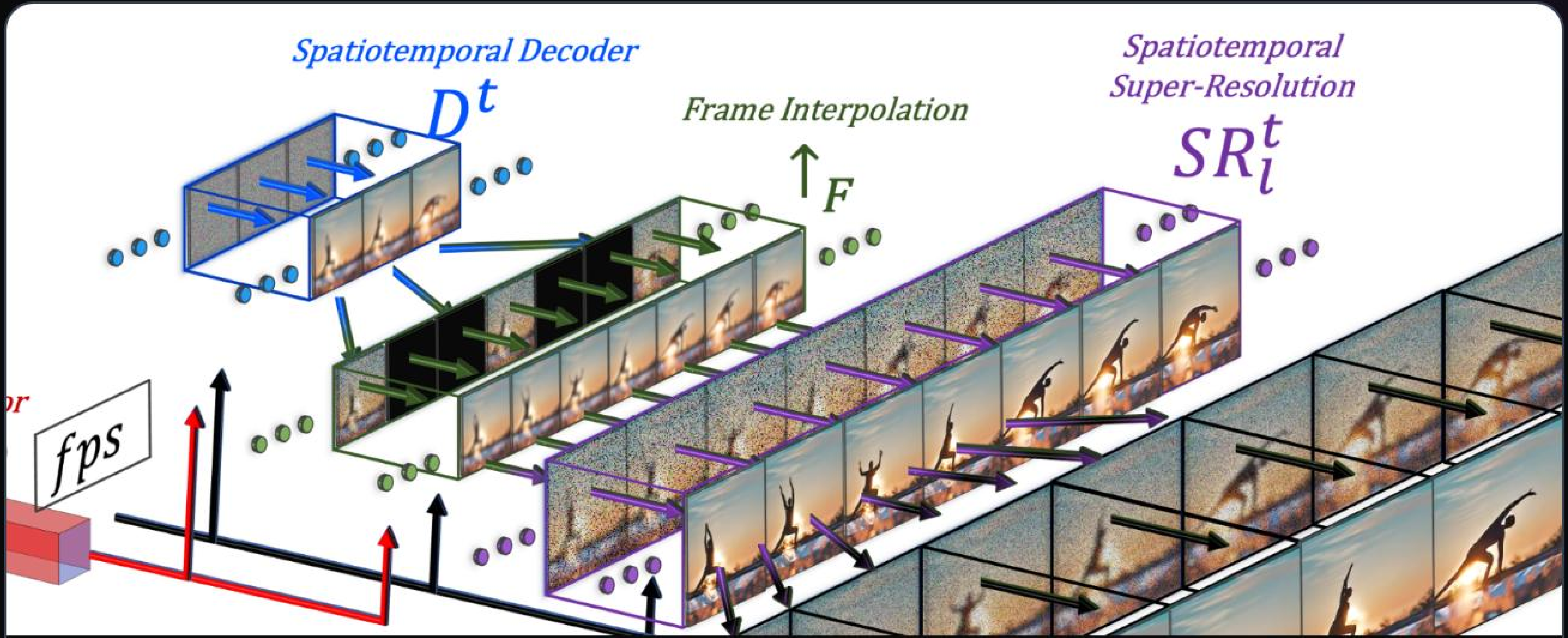
Built upon the powerful 14B parameter Wan2.1-T2V video diffusion architecture.

# Best-in-Class Performance


Vista4D achieves a significant reduction in Fréchet Video Distance (**7.5 FVD** vs 13.0 for GEN3C), marking a new state-of-the-art in reshooting realism.

# Visual Quality Comparison

Raw 4D Point Cloud Render



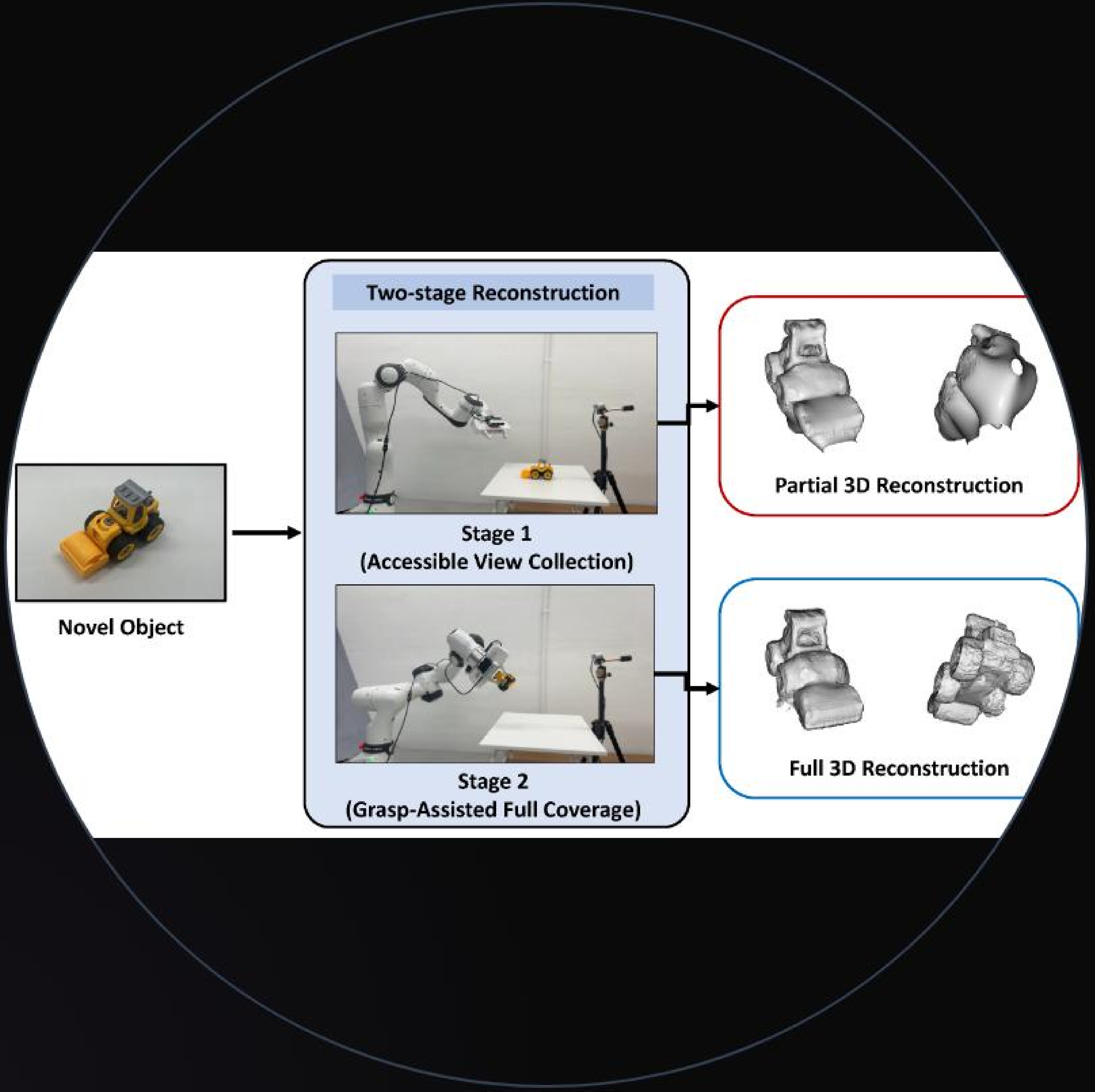

Vista4D Final Synthesis

# Application: Scene Expansion

Vista4D reduces model hallucination by including **static context frames** from casual captures. Users can extended scene geometry beyond the original boundary.

> "The editable 4D context allows for spatial outpainting that remains 3D-consistent across new camera paths."

# 4D Scene Recomposition




## Point Cloud Editing

Because the representation is a readable point cloud, users can **segment and reposition** dynamic objects in 3D space. The diffusion model then synthesizes the edited scene with consistent lighting and motion.

# Questions?

Explore the future of 4D video synthesis at **eyeline-labs.github.io/Vista4D**

arXiv:2604.21915    Eyeline-Labs/Vista4D

# Image Sources

Thumbnail for www.xugj520.cn

https://cdn.hailuoai.com/prod/hlf_images/407568300441010180/images/x0wl0d.jpg

Source: www.xugj520.cn

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https://media.dcrainmaker.com/images/2024/06/Zwift-Ride-DCR.jpg

Source: www.dcrainmaker.com

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https://images.ctfassets.net/et769tc4wc1v/5Ia8aUxoWYIfN8qxCAxKNt/a9bc49f642bd32316fa3bfe594fbbfb2/in-blog-gsvspc-lliverpoolstation.png?w=1920&h=1058&q=50&fm=png&bg=transparent

Source: www.heliguy.com

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Thumbnail for lilianweng.github.io

https://lilianweng.github.io/posts/2024-04-12-diffusion-video/make-a-video.png

Source: lilianweng.github.io

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Source: www.vecteezy.com

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